

you look at it. Ford has already started work on the Adaptive Cruise Control (ACC) which, as we mentioned before, will allow vehicles to maintain safe distances from each other. With ACC, the driver only has to set the speed of his car and the distance he desires to maintain between other vehicles and his car, and if he chooses to he can always over-ride the system by applying the brakes. A microwave radar unit or lasers transceivers fixed in front of the car determine distance and the relative speed of neighboring vehicles. The brakes and the throttle of the car is also handled by the ACC. Regular cruise control systems only track the desired vehicle speed, but ACC is unique in the aspect that it understands and intelligently adapts itself to the behavior of the respective vehicles and using Radar or Lidar it follows what we shall call the 'leader' vehicle at a time gap requested by the driver. When this 'leader' vehicle is not present, the ACC reverts to regular cruise control and defaults to the speed set by the driver. The Nissan Q45 and FX45, the Mercedes S-class, the Lexus 330 and 430, the Audi A8, and select Jaguar and Cadillac models all use ACC and they obtain the required data using forward-looking sensors.

However, ACC also has its drawbacks. It is vulnerable to interference, noise, and false alarms as it isn't yet a perfected system and because of this heavy delays may creep into the system because of which the vehicles ability to follow required vehicle and maintain required speed are dampened. To better this system, there is another version of the ACC called Cooperative ACC. In the CACC, the forward-looking sensors we mentioned before are now coupled with a wireless communication link that connects the leader and the follower vehicle and hence is able to pass information between the two so as to get a better judgement on the next decision which could be increasing speed or decreasing speed or applying the brakes. If this system is commonplace, all vehicles will be connected through this system and so traffic will adjust itself in the most beneficial way for the good of all the vehicles and hence will make our commutes a lot less frustrating.

In addition, the internet is paving the way for enhanced use of embedded technologies. Imagine driving past a petrol pump on your way to Lonavala only to be given a notification that you need a refill (which is pretty awesome), or crossing a toll naka and having the toll amount directly deducted from your bank account (which we just realised is not so awesome). This system is already in place in quite a few parts of the world.